

8

USING “DIGITAL ACADEMIES” TO CLOSE THE SKILLS GAP

by Rubén Mancha and Salvatore Parise

Today, the problem most companies face in executing their digital transformation is not access to technologies but a shortage of workers with digital and data science skills. The demand is growing faster than the supply. In other words, recruitment can only get your company so far. To get the rest of the way, companies need to commit to training and upskilling, giving employees opportunities to learn about technologies—and room to experiment (and fail) with them.

There are proven benefits to internal upskilling efforts, but also limitations. On the benefits side, organizations with more advanced upskilling programs report more significant innovation, digital transformation, productivity, business growth, and employee engagement and retention.¹ However, self-directed learning adds to the individuals’ workload—effort rarely acknowledged or rewarded by the organization. And widely available digital and data science learning platforms offer generic resources not linked to the organizations’ culture, innovation dynamics, and transformation opportunities, which will limit the value extracted from the workers’ gained skills.

Despite the benefits, only 18% of leaders “believe their organization has made ‘significant progress’ in establishing an upskilling program,” according to a survey of CEOs conducted by PwC.² Leaders reported struggling with delimiting the skills and allocating resources for upskilling initiatives, motivating employees, and retaining upskilled workers.

In our research on upskilling efforts, we found that “digital academies” are among the most successful approaches to closing the digital skills gap. A digital academy aims to catalyze how employees interact with digital and data science and lead the transformation of processes, products, and services. As a companywide effort, digital academies meet unique criteria:

- Digital academies are specific to the company’s culture and narrative.
- While diverse in length and content, their programs are highly experiential and considerate of organizational team dynamics.
- They reach across the enterprise to train new employees for technical career paths, knowledge workers on the use of specific technologies, and managers and executives to raise data science awareness and the impact of technologies in different areas of the business.

By focusing on how to use technology in the context of the organization and its digital vision, they help create and reinforce a specific culture around tech and innovation in a way that more generalized online trainings simply can’t.

These have been built at companies such as DuPont, Gestamp, and Deloitte as a central initiative of their digital transformation efforts, resulting in increased capability of using data and adapting to technological advancements. In this article, we explain why digital academies work—and what companies need to keep in mind when designing their own.

DuPont’s Spark Digital Academy

In 2020, DuPont, the multi-industrial specialty products company, implemented a digital transformation plan to become more innovative. To follow through on this ambition, however, the company “recognized that digital transformation was going to require the upskilling of our employees: both educate them on state of the art—what is possible—and change their skills,” said Duncan Coffey, DuPont’s information and data science leader. There was a shortage of the necessary skills among current employees, and competition to recruit people with those skills was stiff.

In response, the company launched Spark Digital Academy in 2021. It aimed to create a bottom-up culture of data, agility, and experimentation across the organization; train workers on the use of data and digital technologies; and signal the company’s priority in digital innovation. The course topics included technology (data science, robotic process automation, internet of things), capabilities (user experience, design thinking, agile/scrum, change management), and digital-specific product topics.

Since 2021, more than 500 DuPont employees have participated in instructor-led courses. The Spark Digital Academy has helped employees adopt digital technologies, optimize processes, and create value with digital products. The positive impacts of DuPont’s upskilling program reinforce the foundation of the organization’s digital transformation across all job roles and levels.

How can your company create a digital academy initiative? DuPont’s efforts, in line with our research, suggest a few fundamental principles that lead to success.

Design Training That Will Serve Multiple Employee Segments

To facilitate the digital transformation of the organization, digital academies need to catalyze a digital and data science culture that works across the organization, helping practitioners, project leaders, and executives. When designing courses, it is important to connect to different target audiences in the organization and learn about their digital needs.

DuPont’s digital academy served to train two primary roles: the “citizen data scientist” and the “translator.”

Citizen data scientists were employees who sought in-depth data science and digital expertise that they could apply directly to their job domains. They needed tools and software and access to professionally trained data scientists so they could build and implement models in their jobs. To address their needs, the academy offered a foundational course for data science, followed by an advanced course for data science. Both courses typically required students to bring a project to do in parallel so they could apply the learnings immediately. Employees usually took them in series.

TABLE 8-1

Possible learning paths in data science for the “citizen data scientist” role

Content	Modality	Duration
Curated tutorials on technologies and digital competencies	Self-paced materials, online	From a few hours to several days
Foundations of data science	Instructor-led course	Six hours per week, six weeks
Advanced data science	Instructor-led course with an experiential team project	Six hours per week, six weeks
Participation in communities of peer mentoring	On-demand	As needed

Translators, on the contrary, were employees who often had enough job experience and an end-to-end, big-picture view of work processes to understand how and where improvements could be made and wanted to bring digital solutions to science, engineering, and business problems. For them, digital and data science training opened a new lens for them to unlock value. The academy offered a data science course for senior leadership and subject business experts, providing general awareness of technologies and data science concepts.

Ensure Experiential Projects Are Part of the Digital Academy

Academies work best when employees bring on-the-job projects with them. This is true for a few reasons:

1. It ties digital principles to a real business problem, need, or opportunity so that essential concepts are not preserved in a vacuum or generic context.
2. Deeper learning occurs at work, not in the classroom.
3. It exposes students to less obvious problems that must be overcome on typical digital projects, such as quality data capture and cleaning.
4. Experiential projects in the classroom are also a way of derisking projects by quickly testing assumptions before spending large amounts of time, effort, and money.
5. Finally, it is a way to communicate value to business leaders so that they see the business value and can further sponsor and fund the effort.

For example, the DuPont digital academy's latest courses on robotic process automation (RPA) included hands-on activities on cloud servers, so students could develop and implement RPA solutions during the course. Work was often done in groups to replicate team dynamics. The academy encouraged people from the same research group to sign up together for instructor-led courses with experiential components. One team project was started during the academy training. After communicating the project value to leadership, the team was then able to pull in extra resources to execute the project at a higher level when the course was completed.

Develop a Continuous Learning Relationship

One of the biggest mistakes we have seen with companies implementing digital academies is not continuing the relationship with employees after they have taken the course. More than simply helping employees acquire new digital skills, these programs create the space to initiate a process of cultural

change. This process is fragile and needs time, frequent revitalization, and sustained access to resources. To ingrain a lasting digital mindset, academies need to develop an ongoing relationship with employee graduates. This should include:

- Providing services and support after each employee graduates through alum communities, mentor-mentee relationships, and technical assistance and infrastructure support, which is critical for digital and data science projects.
- Collecting continuous feedback from past graduates, internal business leaders, external vendors, and thought leaders on what topics should be offered or updated in the academy.
- Forging continued relationships—especially with project leaders and business leaders (or their direct reports) who took a course from the academy—is essential. These leaders often recommend employees or teams to enroll in a particular set of courses.

For example, DuPont's digital academy built communities of practice that could provide peer support on key topic areas, including a mentor-mentee program to support continued development work for those who completed the new RPA course. The academy also relied on input from leaders in DuPont's digital transformation groups—as well as from external instructors and facilitators—on where to grow the academy next. Finally, to raise awareness of the opportunities it offered, it implemented a badging and certificate program to build awareness, recognition, and advocacy. Course graduates received a digital badge that they could display on their email signatures and LinkedIn profiles.

Design the Digital Academy with Flexibility in Mind

In today's hybrid work environment with its increasingly uncertain work demands, companies should design their digital academies with course content and delivery flexibility to gain the highest potential demand. Academies that offer a portfolio of approaches—from self-paced on-demand courses to synchronous virtual or face-to-face courses—often achieve higher performance than those with a strict, narrow delivery format.

Academies should decide on a combination of experiential, peer, and formal instruction that works for the audience and the learning they seek. They should pay attention to self-paced content—not everything has to be instructor-led—and the value of experiential learning tied to the employees' roles. When possible, they should offer cases and content linked to the organization since this content is perceived as being more relevant. Also, employees with little or no experience with digital technologies may benefit from more direct instruction and guided activities. At the same time, advanced groups working on experiential projects may make the most of peer learning and open-ended project work.

Courses should also be designed with intensity and frequency in mind. For example, live courses that are spread out over days or weeks may be more accommodating and welcoming than courses that require several consecutive days of full-day participation. Popular instructor-led courses should also be repeated multiple times over the year.

At DuPont, their digital academy experimented with the delivery format of their data science foundation and advanced courses. Initially, they were delivered in an 8 a.m. to 5 p.m. format over a whole work week. After experimenting and receiving student feedback, a program of four hours per day over two weeks was very successful and better received.

Finally, providing learning pathways for their self-paced courses answered employees' "What course should I take next?" questions. The academy curated 20 self-paced tracks, which have generated thousands of views.

In conclusion, we should see the continued proliferation of corporate digital academies to upskill employees. This will have implications for online course providers such as Coursera and edX, consulting companies, and higher education institutions on how they deliver digital and data science training. For example, more colleges and universities offer specialized degree programs, certificates, and executive education focusing on this domain. Since many corporations interested in upskilling their workforce need more expertise, resources, and bandwidth to do digital academies completely themselves, we predict they will partner with these external institutions to design and deliver this training quickly and at scale.

TAKEAWAYS

Digital academies are among the most successful approaches to closing technological skill gaps. These internal upskilling programs can catalyze how employees interact with data science and lead the transformation of processes, products, and services. As companywide efforts, digital academies meet unique criteria and require new approaches.

- # They are specific to the company's culture and narrative. While diverse in length and content, their programs are highly experiential and considerate of organizational team dynamics.
- # They reach across the enterprise to train new employees for technical career paths, knowledge workers on the use of specific technologies, and managers and executives to raise data science awareness.
- # Many corporations interested in upskilling their workforce will be best served by partnering with external institutions to design and deliver this training quickly and at scale.

Acknowledgments: The authors would like to thank the Innovation Research Interchange for sponsoring the study of DuPont's upskilling efforts and Duncan P. Coffey, information and data science leader; Hai Zhu, digital technology leader; and Kristen Ann Kozub, learning and development analyst, for sharing their insights and experiences launching and scaling DuPont's Spark Digital Academy.

NOTES

1. "Navigating the Rising Tide of Uncertainty," PwC, <https://www.pwc.com/gx/en/ceo-survey/2020/reports/pwc-23rd-global-ceo-survey.pdf>.

2. "Navigating the Rising Tide of Uncertainty."

Adapted from content posted on hbr.org, January 17, 2024 (product #H07Z3X).